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Source: *Child Development*, Jun., 1990, Vol. 61, No. 3 (Jun., 1990), pp. 893-910

Published by: Wiley on behalf of the Society for Research in Child Development

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Juvenile Delinquency and Attention Deficit Disorder: Boys' Developmental Trajectories from Age 3 to Age 15

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MOFFITT, TERRIE E. *Juvenile Delinquency and Attention Deficit Disorder: Boys' Developmental Trajectories from Age 3 to Age 15*. CHILD DEVELOPMENT, 1990, 61, 893–910. This article describes a longitudinal analysis of the behavior of a birth cohort of 435 boys. 4 groups were defined at age 13 on the basis of both self-reported delinquent behavior and professional diagnosis of Attention Deficit Disorder: ADD + delinquent, ADD only, delinquent only, and nondisordered. Biennial correlates of delinquency (antisocial behavior problems, verbal intelligence, reading difficulty, and family adversity) were traced across childhood. The ADD + delinquent boys consistently fared the worst on the assessments of family adversity, verbal intelligence, and reading. Their antisocial behavior began before school age, escalated at school entry, and persisted into adolescence. The ADD-only boys had normal family, intelligence, and reading scores, and showed only mild antisocial behavior in middle childhood. The delinquent-only boys showed no early risk from family, low intelligence, or reading deficit, and remained relatively free of conduct problems until they initiated delinquency at age 13. Persistence of criminal offending beyond adolescence is predicted for the ADD + delinquent boys.

Juvenile delinquency is a diverse phenomenon. Some adolescents behave antisocially as a complication of psychopathology, but many engage in illegal behavior without such disorder. Some adolescent delinquents go on to be adult criminals, but most do not. It is important to delineate possible subtypes among delinquents and to examine their correlates, developmental precursors, and ultimate outcomes. This longitudinal study of 435 boys focuses on groups defined at age 13 both on the basis of juvenile delinquency and diagnosis of Attention Deficit Disorder. Developmental trajectories from ages 3 to 13 were traced for selected correlates of delinquency (childhood antisocial behavior, low verbal intelligence, reading difficulty, and family adversity) in order to explore the developmental precursors of four different forms of adolescent behavior. The groups' delinquent outcomes to age 15 were also compared.

Historically, longitudinal studies of delinquency have had several distinct objec-

tives. First, such studies can establish the extent to which the tendency to behave antisocially is a stable characteristic of certain individuals, across age and situations (Caspi, Elder, & Bem, 1987; Huesmann, Eron, Lefkowitz, & Walder, 1984; Loeber, 1982).

Second, longitudinal research enables us to identify developmental periods during which certain types of antisocial behavior are virtually normative. At these ages, antisocial behavior is not presumptive evidence of psychopathology. For example, aggression is common in preschoolers (Parke & Slaby, 1983), tantrums are at peak prevalence at age 3 (Rutter & Garnezy, 1983), and many, if not most, male adolescents engage in some delinquent behavior (Wilson & Herrnstein, 1985).

Third, longitudinal follow-ups of early behavior disorder reveal the implications of early behavior problems for a child's adolescent and adult adjustment. Prognostic information places the importance of early intervention in perspective. As a case in point, three follow-up studies of hyperactive chil-

This work was supported by USPHS grant 1 R01 MH43746 and MH45070 from the Antisocial and Violent Behavior Branch of the National Institute of Mental Health and a grant from the University of Wisconsin Graduate Research Committee. The Dunedin Multidisciplinary Health and Development Research Unit is directed by Phil A. Silva, and is supported by the Medical Research Council of New Zealand. Appreciation is expressed to Rob McGee, J. C. Anderson, Pat Brasch, the psychometrists who administered the WISC-R, and the police department of Dunedin, New Zealand. Avshalom Caspi, Rolf Loeber, and anonymous reviewers are thanked for their suggestions about the manuscript. Correspondence regarding this article may be addressed to Terrie E. Moffitt, Department of Psychology, University of Wisconsin, 1202 West Johnson Street, Madison, WI 53706.

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dren have now shown that about half of these children are at significant risk for criminality and drug abuse as adults (Gittelman, Man-nuzza, Shenker, & Bonagura, 1985; Satter-field, 1987; Weiss, Hechtman, Milroy, & Perlman, 1985).

Fourth, longitudinal research can be used to identify early life predictors of crime and delinquency that are useful for constructing theories of crime causation (Loeber & Dishion, 1983). In one well-known longitudinal study, Robins (1966, 1979) showed that childhood conduct disorder is an almost necessary precursor of adult antisocial personality disorder (APD), thereby challenging theorists to account for childhood origins of APD.

Fifth, only longitudinal designs can establish the temporal ordering of correlates of antisocial behavior so as to suggest possible causal chains. In addition, such designs can point to links in the chain that should be targets for timely intervention efforts. For example, Hirschi's (1969) social control theory links school failure causally to delinquency. McGee, Share, Moffitt, Williams, and Silva (1988) showed that reading failure precedes onset of many cases of problem behavior. Also, Elliott and Voss (1974) found that, for many delinquents, the rate of self-reported delinquency declined following dropout from school. These longitudinal analyses provide compelling support for the prediction based on control theory that school failure is a proximal cause of delinquency. In addition, the findings suggest that programs that help children succeed academically might also deter delinquency.

Sixth, and finally, Loeber (1988) has argued that we should study developmental pathways that may discriminate different subtypes of offenders. Differential trajectories may distinguish, for example, persistent high-rate perpetrators of serious crimes from temporary or mild offenders. In this vein, early onset of antisocial behavior has been identified as a robust predictor of later violent or serious offending, whereas late onset indicates a better prognosis (see Farrington, 1987, for review).

The present study addresses the research objectives described above in the context of a longitudinal investigation of the relations among cognitive abilities, antisocial behavior, and family adversity in a representative birth cohort of boys. In an earlier cross-sectional study of these same subjects (Moffitt & Silva, 1988a), a small subgroup of delinquents was identified which could be distinguished from other delinquents by a diagnostic history of Attention Deficit Disorder (ADD), poor scores on neuropsychological tests of verbal and auditory memory abilities, relatively greater family adversity, and extreme parent and teacher ratings of aggression. This ADD + delinquent group differed from other boys with ADD on neuropsychological status and on aggressiveness, but the two ADD groups proved to be indistinguishable on parent and teacher ratings of the symptoms essential to their diagnosis of ADD: inattention and hyperactivity. The significant group differences that were obtained were large; the ADD + delinquent group's mean problem scores were between 0.75 and 2.00 SD greater than those of the comparison groups. These findings suggest that neither attention deficit disorder nor adolescent delinquent behavior are related to neuropsychological deficit and aggression in a univariate fashion. Rather, Moffitt and Silva (1988a) argued that it is the specific co-morbidity of ADD and antisocial behavior that warrants further investigation. Indeed, a growing body of literature has documented discriminant validity for the co-morbidity of ADD and antisocial behavior with respect to features such as peer relations, social cognitive information processing, and prognosis (see Hinshaw, 1987, and Milich & Landau, *in press*, for reviews). This article longitudinally examines the developmental histories of the four groups defined earlier by Moffitt and Silva: ADD + delinquents, ADD nondelinquents, non-ADD delinquents, and nondisordered boys.¹

The ADD + delinquent group is especially interesting to criminologists because there is some evidence that this group may be at relatively greater risk for persistent serious criminal offending (i.e., Rutter, Tizard, &

¹ Analysis of categorical groupings, instead of continuously distributed scale data, often brings the disadvantages of arbitrary cut-off scores for group membership and diminished statistical power, among others. However, previous research had suggested a theoretical rationale for the present study's focus on diagnostic categories; failure to isolate the relatively small subgroup with co-morbidity was expected to attenuate the association strength of its correlates. Furthermore, figural plotting of developmental trajectories required a categorical approach. The arbitrariness of group membership cutoffs was addressed by using DSM-III criteria and requiring confirmation from multiple sources. Any loss in statistical power should have had a conservative effect on the ability to report group differences here.

Whitmore, 1970). In a review of longitudinal studies of predictors of later recidivistic serious criminal offending, co-morbidity of ADD and antisocial behavior was purported to be a better predictor than early self-reported delinquency alone (Loeber & Stouthamer-Loeber, 1987). Farrington, Loeber, and van Kammen (in press) found a strong interactive predictive effect between problems of "hyperactivity, impulsivity, and attention" and childhood conduct disorder (CD) on later criminal offending. Whereas 12.6% of nondisordered boys were convicted, conviction rates for boys with ADD only, CD only, and both CD and ADD were, respectively, 23.5%, 35.0%, and 45.8%. Similar predictive results that are specific to ADD + CD co-morbidity have been reported by Lambert (1988) for delinquency outcome at age 17, and by Loney, Whaley-Klahn, Kosier, and Conboy (1983) and Magnusson (1988) for adult criminal offending.

In combination, these studies suggest that children who have the combination of ADD and early antisocial behavior problems may constitute a subgroup that is at significant risk for sustaining their antisocial behavior through adolescence and thereafter emerging into adult lawbreakers. Now that the future implications of this co-morbidity are beginning to be understood, it is timely to examine the developmental histories of subjects with both ADD and conduct problems.

The research reported here addresses six questions about the subjects, earlier identified by Moffitt and Silva (1988a), who had adolescent delinquent behavior at age 13 as well as an earlier diagnosis of ADD. (1) How early in life can the ADD + delinquent group be differentiated from relevant comparison groups on the basis of behavior and context? (2) Is the ADD + delinquency group more consistently antisocial across childhood than the comparison groups? (3) Can distinct temporal sequences of the correlates of ADD + delinquent co-morbidity (cognitive deficit, reading difficulty, family adversity, antisocial childhood behavior) be identified so as to suggest causal chains? (4) Can correlates of ADD + delinquency be fitted to conceptual models of risk and resilience? (5) Does the ADD + delinquency group differ from comparison groups on correlates of cognitive deficit, such as low birthweight or poor preschool motor abilities, which might suggest a neurological basis for the co-morbidity? (6) Is this group more delinquent 2 years later, at age 15, than the comparison groups?

Method

Subjects

Subjects were adolescent boys involved in the Dunedin (New Zealand) Multidisciplinary Health and Development Study. The cohort's history has been described by McGee and Silva (1982). Briefly, the study is a longitudinal investigation of the health, development, and behavior of a complete cohort of consecutive births between April 1, 1972, and March 31, 1973, in Dunedin, New Zealand. Perinatal data were obtained and, when the children were traced for follow-up at 3 years of age, 1,139 children were deemed eligible for inclusion in the longitudinal study by residence in the province. Of these, 1,037 (91%) were assessed. Since then, follow-ups have been conducted every 2 years. Data were collected for 991 subjects at age 5, 954 at age 7, 955 at age 9, 925 at age 11, 850 at age 13, and 995 at age 15. McGee (1985) has compared children who were lost to the study at each age with those remaining by age 11 and found no systematic differences in social class, IQ, or a variety of behavioral variables. When compared to the New Zealand general population, the cohort is slightly biased toward higher social class levels: on a six-level social class scale (Elley & Irving, 1972), 7% of New Zealand males but 11.7% of cohort fathers were rated in the highest level. It is predominantly of European ancestry (less than 2% Polynesian) and is therefore comparable to similar white samples from other English-speaking Western cultures.

Variables

Antisocial behavior.—At ages 5, 7, 9, 11, and 13, a parent and a teacher rating of each subject's antisocial behavior were standardized and averaged to form a composite antisocial score. At ages 5, 7, 9, and 11, the ratings were made using the "antisocial" subscale of the Rutter Child Scales (RCS; Rutter et al., 1970). Forms A and B of the RCS are designed to be filled out by parents and teachers, respectively, and the 12 items describing antisocial behaviors were coded by them as *does not apply* (0), *applies somewhat* (1), or *certainly applies* (2). At age 13 the teachers again filled out the RCS-B, but the parent filled out the Revised Behavior Problem Checklist (RBPC; Quay & Peterson, 1987), which contains more extensive and age-appropriate items than the Rutter scale. Items were scored 0, 1, or 2, as they were for the RCS. The Socialized Aggression (SA) subscale of the RBPC was used in combination with the teachers' RCS-B rating to compute the age 13 composite antisocial score. Corre-

lations between the parent and teacher ratings at ages 5, 7, 9, 11, and 13 were .02, .22, .23, .36, and .47, respectively.

Family adversity.—A measure of factors related to social class was calculated for each data collection wave (ages 3, 5, 7, 9, 11, and 13). The measure was based on the model of Rutter's adversity index (Rutter, 1978). It was calculated by adding 1 point for each of the following: low parental education (less than high school graduation), low parental income (under \$10,000 per year), semiskilled or unskilled parental occupation, teen-aged mother, single-parent status, separation from parents for at least 1 year, large family size (five or more children), poor maternal mental health (below the sample tenth percentile on the "Malaise Inventory" of Rutter et al., 1970), low maternal IQ (below the sample tenth percentile, Thurstone & Thurstone, 1973), and a poor score on a measure of family social environment (below the tenth percentile on the Moos Family Relations Index, Moos, 1974). Because all of these variables were not available for every wave, all of the adversity indexes could not be calculated identically. Nevertheless, at least six of the 10 variables were included in the index at each assessment age. (Teen mother and maternal IQ were used only at the age when they were assessed.)

Verbal cognitive ability.—Previous findings of verbal deficits among delinquents have been more robust than findings of deficits in other mental functions (see Moffitt, in press, for review). For this reason, measures of verbal abilities were used here. At ages 7, 9, 11, and 13, the Wechsler Intelligence Scale for Children—Revised (WISC-R; Wechsler, 1974) was administered by trained psychometrists according to the standard protocol, with two exceptions: (1) three subtests (Comprehension, Digit Span, and Picture Arrangement) were omitted because of time constraints, and (2) six Information items have been modified in the New Zealand version in order to insure local relevance (e.g., "Who discovered America?" was changed to "Who discovered New Zealand?"). Verbal IQ scores were prorated following the procedure in the test manual, using four subtests. At age 5, the Stanford Binet Intelligence Scale (Terman & Merrill, 1960) was administered and scored according to standard protocol. It served as the age 5 intelligence measure.

Reading achievement.—At ages 7, 9, 11, and 13 the subjects took the Burt Word Reading Test (Scottish Council for Research in

Education, 1976). This is a word-recognition reading test normed for New Zealand children. In format it resembles the American Wide-Range Achievement Test of reading.

Motor functions.—Motor coordination was assessed at age 3 by trained pediatricians using the 28 most difficult motor items from the Bayley Scales of Infant Development (Bayley, 1969), and at age 5 using the motor scales from the McCarthy Scales of Children's Abilities (McCarthy, 1972). The motor measures were examined here as indices of central nervous system integrity, to complement the intelligence tests of cognitive ability.

Perinatal problems.—The subjects were medically examined, and their medical records were obtained, shortly after birth. A scale of overall perinatal difficulties was constructed by adding 1 point for each of the following: low Apgar scores (bottom 10%), pre-term gestational age, low birthweight (bottom 10%), twin birth, and idiopathic respiratory distress syndrome. At least one perinatal problem was experienced by 46% of the sample, but only 19.4% had two or more problems. Earlier research with this sample (Silva, McGee, & Williams, 1984a, 1984b) has shown that low birthweight was more closely associated with low IQ and behavior problems than were the other perinatal factors, so this variable was also examined independently here. Only 80 infants weighed less than 2,500 grams, and only three weighed less than 1,500 grams, so that severe low birthweight was rare. In the Dunedin sample, neither perinatal problems nor birthweight were significantly correlated with parental SES (Silva et al., 1984a).

Delinquency.—The Self-Reported Early Delinquency Instrument (SRED, described fully in Moffitt & Silva, 1988b) was administered to the subjects at age 13 by an examiner who was blind to their scores on other variables. This measure was designed specifically for use in New Zealand, and it includes validity checks between card-sort and interview protocols, screening and assistance for poor readers, and a score weighted for item seriousness. The SRED contains 29 items tapping "norm-violating" behaviors and 29 items tapping more serious illegal behaviors. The illegal items can be divided into subscales representing vandalism, aggression, theft, and controlled-substance offenses. At age 15, the 29-item illegal scale was readministered to all subjects.

One-month test-retest reliability ($r = .85$), internal consistency (Kuder-Richardson

Formula 20, $r = .90$), and concurrent validity (with parental report of subjects' socialized aggressive behaviors, $r = .43$, $p < .001$) were assessed at age 13 and found to be adequate for research. The subjects with police records at age 13 averaged 11.89 self-reported acts, as compared to 5.24 for subjects not known to police, thereby demonstrating good criterion validity. Thefts (especially shoplifting) accounted for 41.2% of acts reported by the subjects at 13, minor assault for 24.7%, and vandalism and substance abuse for 10.7% and 9.9%, respectively. These relative rates were very similar to the arrest rates for crime types in the Dunedin police district. The sample was representative of official delinquency in the community; 6.1% of the sample and 6.2% of district residents who were born in the same year had police records by their thirteenth birthdays.

History of attention deficit disorder.—Forty-five boys and eight girls with ADD were diagnosed at age 11 as follows (the diagnostic procedures have been described in detail by Anderson, Williams, McGee, & Silva, 1987). Three sources existed for report of symptoms meeting DSM-III criteria for ADD. The child reported symptoms during an interview with a child psychiatrist who used the Diagnostic Interview Schedule for Children—Child Version (DISC-C; Costello, Edelbrock, Kalas, Kessler, & Klaric, 1982), and each child's parent and teacher filled in the inattention and motor hyperactivity subscales of the Rutter Child Scales A and B (Rutter et al., 1970; the Rutter items were supplemented so that full DSM-III criteria for ADD-H could be met). In 35 cases, two of the three possible reporters provided independent consensus report of all DSM-III criterion symptoms, and in 10 cases one reporter met the full criterion and another met some criteria, thereby ensuring pervasiveness of the symptoms (Schachar, Rutter, & Smith, 1981). In an additional eight cases, one reporter alone provided all symptoms to meet diagnostic criteria. Data concerning problem behavior collected from parents and teachers when the children were 5 and 7 years old confirmed onset of the disorder before age 7 for all 53 cases. At age 11, the prevalence of ADD so diagnosed was 6.7%, the sex ratio was 5.1 boys to 1 girl, and 85% of the cases had more than two symptoms of hyperactivity in addition to their attentional problems.

Identification of Groups

At age 13, a group of subjects most heavily involved in delinquent activities was identified by exploiting the possibility for

agreement between four available sources of information about antisocial behavior. The advantages of this approach have been described by Loeber and Dishion (1983). Subjects could report themselves delinquent by scoring above the eighty-fifth percentile on the SRED. (Use of the eighty-fifth percentile cutoff was suggested by an observed discontinuity in the skew at this point in the distribution.) Parents could report their children delinquent by providing scores above the eighty-fifth percentile on the Socialized Aggression subscale of the Quay and Peterson Behavior Problem Checklist—Revised (BPCR; Quay & Peterson, 1987), which was completed at the time of the age 13 assessment. Teachers could report a child delinquent by giving him or her antisocial subscale scores beyond the eighty-fifth percentile on the Rutter Child Scale B (RCSB; Rutter et al., 1970), also completed when the subjects were 13 years old. All percentile ranks were calculated separately for boys and girls, so that group membership reflected serious delinquency *relative to same-sex peers*. A subject was assigned to the delinquent group if at least two reports met the above criteria, or if a file was found for him or her in the police district office. Thus, subjects could not name themselves delinquent without the consensus of at least one adult reporter. This reduced the chance of erroneous group assignment resulting from overzealous self-report. Likewise, a subject who underreported delinquent acts could nevertheless be designated delinquent by agreement between adult reporters. (Subjects who were rated among the top 15% by only one source remained in the nondisordered comparison group, a practice that should exert a conservative influence on our ability to detect group differences.)

Seventy-one boys and 53 girls met these criteria. Because the prevalence of ADD was so low for girls (1.2%), statistical analysis of gender $\times$ group differences would have been unreliable (e.g., only three girls had both ADD and delinquency). Therefore, only the boys were used for this research. At age 13, 435 boys were interviewed and could be classified. The delinquent and nondelinquent groups' mean scores for the SRED were, respectively, 15.73 and 5.70.

A previous childhood diagnosis of ADD was found for 26.8% of the delinquent boys, contrasted with a 4.4% prevalence of childhood ADD among the nondelinquent boys ($\chi^2 = 40.16$, $p < .001$). Indeed, of 35 cases of childhood ADD in the portion of the cohort studied here, 54.3% were found in the delin-

quent group at age 13. The boys were grouped according to their delinquency and ADD designations as follows:² nondisordered, $n = 348$; ADD nondelinquents, $n = 16$; non-ADD delinquents, $n = 52$; ADD + delinquents, $n = 19$.

The four groups' mean scores at the age of 13 have been reported elsewhere for a number of cross-sectional measures (Moffitt & Henry, 1989; Moffitt & Silva, 1988a). Briefly, ADD + delinquents had significantly worse scores than the other three groups on scales measuring IQ, SES, impulsivity, aggression, family adversity, and reading achievement, and on neuropsychological tests of verbal ability and memory, but they were not more hyperactive or inattentive than the other ADD boys, nor more delinquent than the other delinquents. Parents were asked whether they had sought help for their child's behavior problems; affirmative answers were obtained for 48% of ADD + delinquents, 32% of non-ADD delinquents, 13% of ADD nondelinquents, and 8% of nondisordered controls. Medication history is not known for these boys, but pharmacologic treatment of ADD-H is unusual in New Zealand.

Attrition effects.—In order to assess effects of attrition for generalizability of findings, the 435 boys³ studied here were compared to the 101 cohort boys who were not studied here because they had not participated in the age 13 data collection. The earliest available measures were used so as to include as many of the missing boys as possible before they had left the sample at later waves. The studied and not-studied boys did not differ significantly on prevalence of ADD (8% and 9%, respectively, $p > .05$), on age 3 family adversity ($t[534] = 0.66$), on age 5 antisocial behavior ($t[510] = 1.03$), or on age 7 reading ($t[485] = 1.21$). The not-studied boys' mean age 5 IQ score was 5 points lower than the studied boys' mean IQ score ($t[508] = 2.34$, $p = .02$). This IQ difference may be partially explained because institutionalized children who were intellectually handicapped ($n = 6$) were among those not fol-

lowed up at age 13 and therefore not included in this study.

Results

Stability of measures across childhood.—In order to estimate the stability of the measures over the years of the study, a Pearson correlation matrix was calculated for each of the four longitudinal sets of indices: verbal ability, reading achievement, family adversity, and antisocial behavior. For the five indices of verbal ability (i.e., at ages 5, 7, 9, 11, and 13), the correlation coefficients ranged from .61 to .83, with a mean of .71. For the four reading scores, correlations ranged from .72 to .91, averaging .83. These high correlations represented relatively stable performance on IQ and reading tests across childhood for the cohort. Correlations among the six adversity indexes ranged from .28 to .92, averaging .59, suggesting moderate fluctuation in families' positions on the adversity index over the 10 years. The correlations for the five indices of antisocial behavior ranged from .20 to .63, averaging .39, indicating relatively less stability in behavior problems from age 5 to age 13. In general, correlations between indices taken at adjacent data waves were largest. All correlations were positive and significant, $p < .001$.

Intercorrelations among dependent variables.—Calculation of the intercorrelations among the dependent variables (within each age of measurement) yielded 27 correlation coefficients, 22 of which were between .25 and .36, suggesting modest intercorrelation among dependent measures. The five exceptions were the relatively large correlations between reading and verbal IQ, which were near .60 at all four ages when reading was tested, and the small correlation between family disadvantage and antisocial behavior at age 13, which was .19. Despite the large correlation between IQ and reading, the variables were judged not to be entirely redundant, and reading data were analyzed separately so as to examine reading as a mechanism through which both low IQ and

² The numbers of boys per group differ from those reported in Moffitt and Silva (1988a) and Moffitt and Henry (1989) because it was necessary to exclude subjects who had not been neuropsychologically tested from those studies, whereas they were included here.

³ Degrees of freedom vary somewhat for analyses because of differing amounts of missing data at each age wave and for each variable. However, the groups were classified using boys available for study at age 13, which was the wave with the lowest participation rate. Therefore, most subjects did have present data for all variables from that, and other, waves. The smallest number of present cases for any individual variable in this study was 417 (96%) for the age 7 measure of reading achievement. Listwise deletion of cases with missing data yielded smaller numbers for some multivariate analyses.

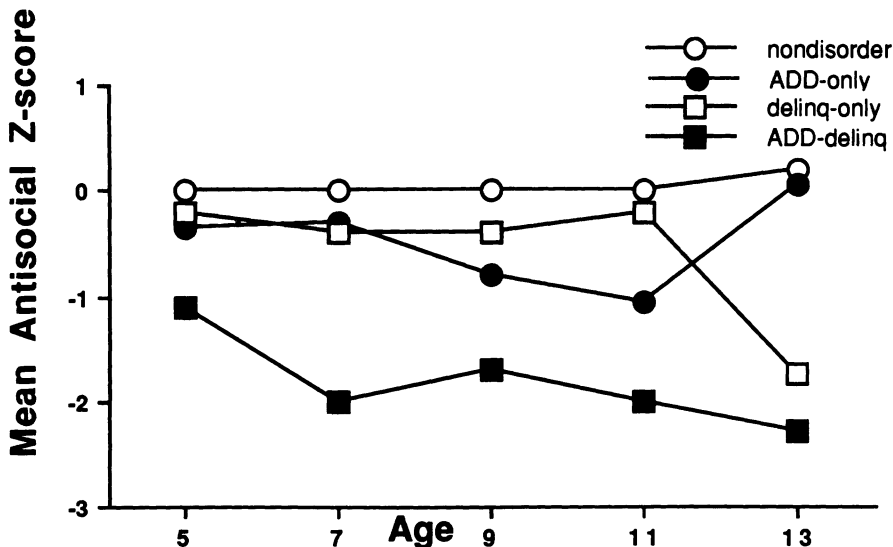


FIG. 1.—Mean standardized scores on a measure of antisocial behavior taken at five successive ages, as a function of delinquency and attention deficit disorder. Negative scores indicate greater levels of antisocial behavior.

early behavior problems might influence delinquent outcome.

Developmental analyses.—Figures 1–4 illustrate the relative positions of the four groups’ mean scores at each age for antisocial behavior, family adversity, verbal ability, and reading. Scores were standardized as Z scores (with a sample mean of 0 and SD of 1) before constructing the figures, so that equivalent

metrics at each age would allow group comparison across ages. For expository purposes, scores were reflected such that negative scores consistently indicate relatively worse status in all four figures, that is, low scores indicate greater family adversity, poorer verbal and reading scores, and more severe antisocial behavior. Group main effects and group $\times$ age interaction effects for each variable were tested using repeated-measures multi-

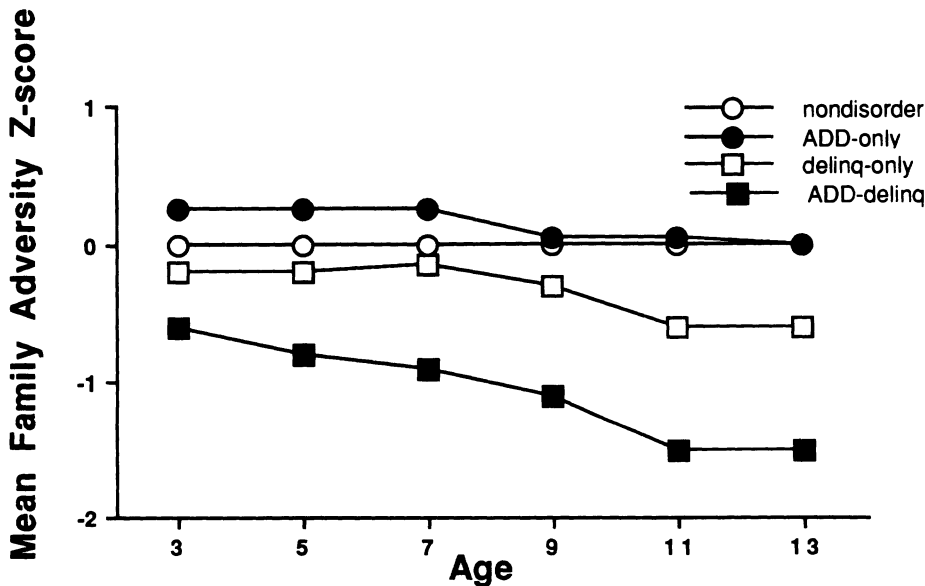


FIG. 2.—Mean standardized scores on a measure of family adversity taken at six successive ages, as a function of delinquency and attention deficit disorder. Negative scores indicate worse adversity.

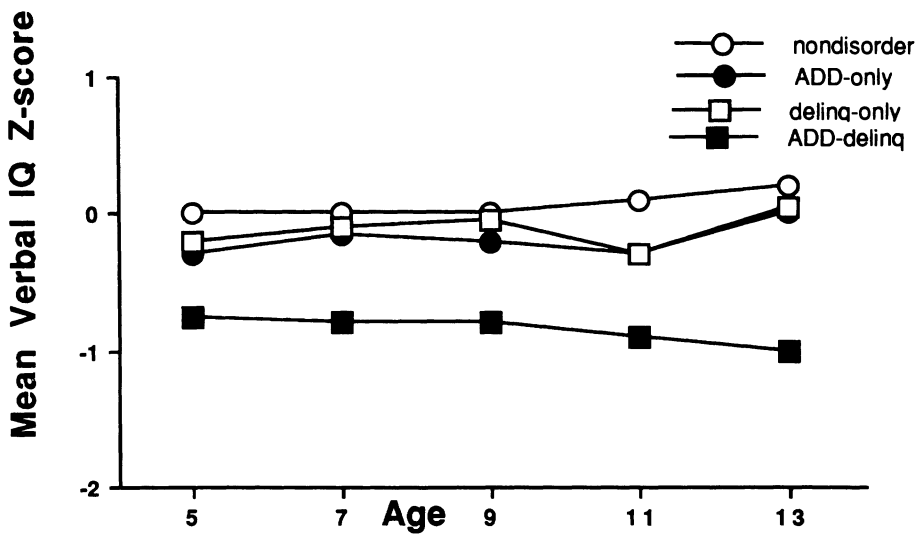


FIG. 3.—Mean standardized scores on measures of verbal intelligence taken at five successive ages, as a function of delinquency and attention deficit disorder. Negative scores indicate poorer verbal ability.

ple analysis of variance (MANOVA). Because of the large number of effect tests conducted here, the exact p values are reported (except where $p < .001$, or for Scheffé's tests, which required $p < .05$) so that readers may evaluate the risk of Type I error. Note that the group sizes were similar for the two ADD groups, so that a significant difference that is reported between nondisordered controls and ADD + delinquents, but that was not found between controls and ADD nondelinquents, is not likely to be the result of differential statistical

power for the Scheffé's comparisons with the nondisordered group.

These analyses address the first three research questions: (1) How early in life can the ADD + delinquent group be differentiated from comparison groups? (2) Is the ADD + delinquency group more consistently antisocial than comparison groups across childhood? (3) Can distinct temporal sequences of the correlates of ADD + delinquent comorbidity be identified?

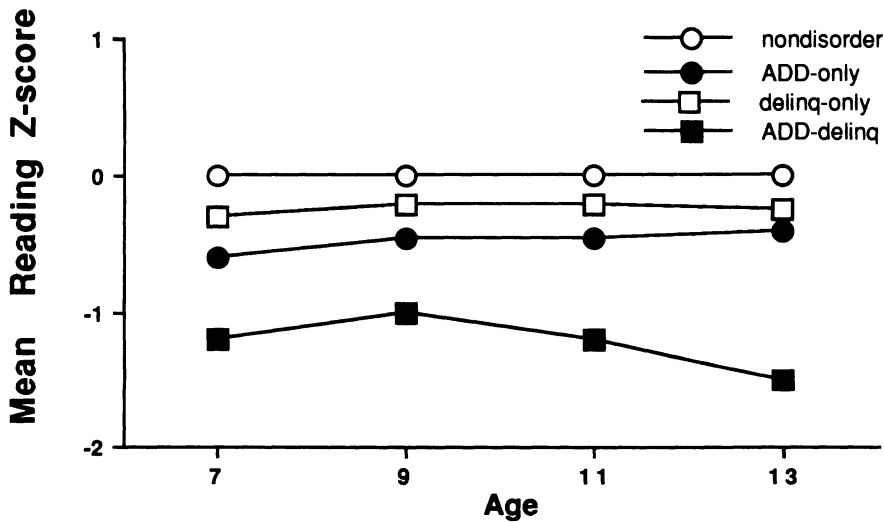


FIG. 4.—Mean standardized scores on a measure of reading achievement taken at four successive ages, as a function of delinquency and attention deficit disorder. Negative scores indicate less reading achievement.

For antisocial behavior, the main effect of group, $F(3,405) = 57.75$, $p < .001$, and the interaction of group with age, $F(12,1202) = 14.92$, $p < .001$, were markedly significant (see Fig. 1). Examination of the post hoc Scheffé's tests revealed differences between the groups in the developmental patterns of their antisocial behavior. The ADD + delinquents were rated as significantly more antisocial than nondisordered controls at every age (and more antisocial than other delinquents at four of the five ages). Decline in their mean scores across age suggested a trajectory of deterioration in their behavior between age 5, when they scored 1.1 SD below the sample mean, and age 13, when they scored 2.3 SD below the mean, as illustrated in Figure 1. By contrast, the antisocial behavior of the ADD boys who had not been delinquent at 13 was much milder and was also more variable across childhood. This pure ADD group exhibited notable amounts of antisocial behavior only in middle childhood; this group of boys was rated as significantly more antisocial than nondisordered boys at ages 9 and 11, and as more antisocial than non-ADD delinquents at age 11. The non-ADD delinquents showed yet another age-related pattern. They were relatively free of antisocial behavior until age 13, when they emerged as significantly more antisocial than nondisordered and ADD-only boys. Indeed, at age 13 their mean antisocial behavior rating was not significantly different from that of the ADD + delinquent group (see Fig. 1). Figure 1 shows that very similar delinquent presentations in adolescence were reached by two distinct pathways. Boys with histories of ADD showed antisocial behavior before school entry, with an exacerbation to more severe levels of antisocial behavior soon after they entered primary school, and they subsequently remained antisocial until high school. By contrast, delinquent boys without ADD began to behave antisocially for the first time at about the age of puberty and high-school entry.

When family adversity scores were analyzed (see Fig. 2), the main effect of group was significant, $F(3,431) = 11.98$, $p < .001$, but the group $\times$ age interaction effect was not significant ($p > .4$). Scheffé's tests showed that ADD + delinquents had significantly ($p < .05$) more adverse families than nondisordered controls at all ages, than other delinquents at ages 9 and 11, and than other ADD cases at ages 5 through 13. ADD nondelinquents' families were slightly less adverse on average than even nondisordered boys'

families at every age, although not significantly so. Although the group $\times$ age interaction term was not significant, family adversity for the delinquent boys did seem to grow slightly worse over time. Non-ADD delinquents' family adversity scores became significantly worse than nondisordered controls' scores only at early adolescence: ages 11 and 13. That change could partially reflect an increase in family distress *consequent* to the emergence of problem behavior at adolescence (Bell & Chapman, 1986). The Family Adversity Index reported in Figure 1 did include measures of mother's mental distress and of family relations, measures that may be expected to react to escalating parent/teen conflict.

The main effect of group upon verbal ability, shown in Figure 3, was significant, $F(4,394) = 2.78$, $p = .02$. The interaction of group with age was not significant ($p > .5$), reflecting the stability of the group differences in tested verbal ability. ADD + delinquents had mean verbal IQs that were significantly lower than nondisordered controls' IQs at all ages, than other delinquents' at 11 and 13, and than other ADD cases at 13. The ADD-only, delinquent-only, and nondisordered groups did not differ significantly on verbal ability at any age; Figure 3 shows that all three of these groups remained within normal limits across childhood.

Results for reading were similar to those for verbal intelligence (see Fig. 4). There was a group main effect, $F(3,393) = 10.57$, $p < .001$, but the group $\times$ age interaction did not attain significance, $p = .06$. Scheffé's tests showed that ADD + delinquents were worse readers than nondisordered boys at all ages. They also read more poorly than other delinquents at 7, 11, and 13, and more poorly than other ADD boys at age 13. Figure 4 shows that they fell even further behind their peers on reading at age 13, when most entered high school. Again, the ADD-only, delinquent-only, and nondisordered groups did not differ significantly in reading at any age, although the ADD-only group consistently scored slightly below the other two groups.

Correlates of group differences.—This section addresses the fourth research question: Does the ADD + delinquency group differ from comparison groups on factors that might suggest a nervous system basis for the co-morbidity? Three indices were examined: low birthweight, number of perinatal problems, and motor abilities (see Table 1). The group differences for birthweight and peri-

TABLE 1
MEANS AND STANDARD DEVIATIONS OF STANDARDIZED (Z) SCORES FOR DESCRIPTORS^a
OF NONDELINQUENTS AND DELINQUENTS, WITH AND WITHOUT HISTORIES
OF ATTENTION DEFICIT DISORDER^b

MEASURE	GROUP			
	Nondisordered	ADD Nondelinquent	Non-ADD Delinquent	Add + Delinquent
Perinatal problems:				
M	.00	.19	– .01	– .28
SD	.97	.97	1.13	1.13
Birthweight:				
M	– .03	– .27	.25	.25
SD	.98	.73	1.05	1.13
Age 3 Bayley Motor Scale: ^{c,d}				
M	– .03	.07	.11	– .70
SD	1.05	.85	.74	1.40
Age 5 McCarthy Motor Scale: ^{c,d}				
M	– .07	– .35	.04	– 1.02
SD	1.00	.98	1.02	.94

^a All variables were (1) standardized and (2) reflected so that negative scores indicate undesirable status: worse perinatal problems, lower birthweight, and poorer motor coordination.

^b Group n's were: nondisordered, 348; ADD nondelinquent, 16; non-ADD delinquent, 52; ADD + delinquent, 19.

^c Scheffé test indicated difference between delinquents with and without ADD was significant ($p < .05$).

^d Scheffé test indicated difference between ADD + delinquents and nondisordered boys was significant ($p < .05$).

natal problems were not statistically significant, although the ADD + delinquency group did have the worst mean perinatal problem scores, as predicted.

The four groups' performances on childhood tests of motor ability were significantly different, and in the predicted direction (see Table 1). On the Bayley scale at age 3, the ADD + delinquent boys had scored significantly (at least .70 SD) below the other three groups, $F(3,432) = 3.02$, $p = .02$, who did not differ significantly from each other. And, on the McCarthy motor scale at age 5, the ADD + delinquent boys again scored significantly (at least .65 SD, $F[3,402] = 5.44$, $p < .001$) below the other three groups, who did not differ significantly from each other. Because the two ADD groups had not differed on parent ratings of their hyperactive behavior at age 5 ($p > .05$), it is unlikely that the distinctively poor performances of the ADD + delinquent group on the two motor abilities tests were solely the result of disruptive hyperactive behaviors during testing.

Additional co-morbid cases.—Although ADD was the most prevalent disorder among delinquents, it was necessary to examine the co-morbidity of delinquency with other behavior disorders before concluding that the combination of ADD and delinquency was somehow unique. Anxiety and depressive disorders were diagnosed in this sample using

the same procedures followed for the ADD diagnoses (see Frost, Moffitt, & McGee, 1989). Only six (11%) of the non-ADD delinquents had depression (2), anxiety (2), or both (2). Because these cases were few, their data were not plotted in Figures 1–4. Briefly, their trajectories for antisocial behavior, verbal IQ, and reading most closely resembled those of the non-ADD delinquents. All of their mean scores were higher than the ADD + delinquents' scores: by .39 to 1.87 SD (median SD = .84) across ages for the behavior, IQ, and reading measures. Only on family adversity did delinquents with a depression/anxiety co-morbidity resemble the delinquents with an ADD co-morbidity (range of mean adversity Z scores = $-.65$ to -1.03). The mean scores on motor and perinatal variables for the six delinquents with depression/anxiety were not significantly different from those of nondisordered boys. No ADD nondelinquent boy had a co-diagnosis; two of the ADD + delinquent boys had anxiety disorders.

A model of vulnerability for ADD + delinquents.—Question 5 was: Can correlates of ADD + delinquency be fitted to conceptual models of risk and invulnerability? Figures 1–4 illustrated that subjects with similar clinical presentations (of ADD-H, or of delinquency) followed very different developmental trajectories, and that these trajectories covaried with individual differences in verbal

TABLE 2
RESULTS OF THE HIERARCHICAL MULTIPLE REGRESSION ANALYSIS FOR THE PREDICTION OF 435 BOYS' ANTISOCIAL BEHAVIOR RATINGS AT AGE 11

Model: age 11 antisocial behavior = B_1 age 5 antisocial behavior + B_2 verbal ability + B_3 family adversity + B_4 age 5 ADD-H + B_5 ADD-H $\times$ verbal ability + B_6 ADD-H $\times$ family adversity + B_7 verbal ability $\times$ family adversity + B_8 ADD-H $\times$ verbal ability $\times$ family adversity + D^a

Step, and Independent Variable	R^2	R^2 Change	F^b	Significance ^b of Increment in R^2	Overall Significance Level
1. Antisocial at 5	.09	...	44.03	.001	.001
2. Verbal	.15	.06	35.82	.001	.001
3. Family	.18	.03	16.97	.001	.001
4. ADD-H at 5	.28	.10	61.20	.001	.001
5. ADD-H $\times$ verbal	.29	.01 ^c	5.36	.021	.001
6. ADD-H $\times$ family	.29	.01 ^c	4.02	.045	.001
7. Verbal $\times$ family	.29	.01 ^c	3.42	.065	.001
8. ADD-H $\times$ verbal $\times$ family	.29	.00	.91	.339	.001
Cumulative multiple $R =$	.54				

^a Where B 's are regression weights and D is a constant.
^b Indicates the significance level for the increment in R^2 at each step.
^c The interaction terms were actually tested one at a time in separate models. Collinearity prohibited entering several multiplicative interaction terms jointly. Shown are t tests of the interaction effects and the independent increase in R^2 associated with each interaction term.

abilities and family adversity. Following the example of Garnezy, Masten, and Tellegen (1984), a multivariate model representing hypothetical interactions between ADD, verbal ability, and family adversity was constructed and tested. Garnezy et al. construed personal qualities (e.g., ADD-H and verbal ability) as imparting vulnerability or immunity to the effects of stressors (e.g., family adversity). Thus, they posited a conditional (interactional) relation between personal qualities and stressors in producing psychopathology. For example, family adversity may amplify the risk effect of ADD-H, with consequent antisocial behavior. In the absence of family adversity, ADD-H's inherent risk may not be fulfilled.

The model shown in Table 2 was tested using hierarchical multiple regression analysis. Antisocial behavior at age 11 served as the dependent variable because it was the latest age at which the ADD + delinquent group alone showed extreme antisocial behavior. (The goal was to test prediction of the subtype of antisocial boy who was characterized by co-morbidity; because the non-ADD delinquents constituted a numeric majority of antisocials at age 13, prediction at that age would have been drowned by the "noise" of their behavior.)

Antisocial behavior at age 5 was entered into analysis first in order to determine whether the other variables could improve

prediction of age 11 behavior beyond mere knowledge of the subjects' earlier antisocial behavior. Verbal ability and family adversity were entered after age 5 antisocial behavior. The measure of verbal ability for this analysis was constructed as the average of the verbal IQ scores across ages 5 to 11. Likewise, the family adversity indices shown in Figure 2 were averaged across ages 3 to 11 to form a summary measure of experienced adversity.

Age 5 ADD-H was entered as the last main effect so as to test whether ADD-H can explain variance in antisocial outcome beyond that explained by the more well-known correlates (early antisocial behavior, verbal ability, family adversity). ADD-H was operationalized as the averaged standardized ratings of hyperactive symptoms made by each child's parent and teacher at age 5 (ratings of attention per se were not collected prior to age 9). All main effects took precedence over interaction terms. Table 2 presents the results of the analyses.

Altogether, 29% of the variance in age 11 antisocial behavior was explained by the model. As predicted, the child's own preschool antisocial behavior was a relatively strong predictor, accounting for 9% of the variance. Nonetheless, each of the other modeled main effects and interaction terms did account for a significant proportion of variance beyond the initial amount explained by

TABLE 3

MEANS AND STANDARD DEVIATION OF STANDARDIZED (Z) SRED^a SCORES FOR NONDELINQUENTS AND DELINQUENTS, WITH AND WITHOUT HISTORIES OF ATTENTION DEFICIT DISORDER^b

Measure	Nondisordered	ADD nondelinquent	Non-ADD Delinquent	ADD Delinquent
Total SRED variety score:				
M	-.10	-.08	.75	.90
SD	.89	.41	1.66	1.76
Aggressive acts: ^c				
M	-.16	-.19	.70	1.91
SD	.71	.82	1.43	2.30
Thefts:				
M	-.17	.23	.83	1.35
SD	.67	1.28	1.58	2.06
Drug and alcohol use:				
M	-.13	.12	.73	.70
SD	.48	.97	1.96	2.83
Destructive vandalism:				
M	-.16	-.21	.82	1.30
SD	.76	.46	1.46	1.84

^a Higher scores indicate more self-reported illegal acts. Items were weighted for seriousness before computing the four subscales.

^b Group *n*'s were: nondisordered, 348; ADD nondelinquent, 16; Non-ADD delinquent, 52; ADD + delinquent, 19.

^c Post hoc Scheffé Test indicates difference between delinquents with and without ADD was significant ($p < .05$).

early antisocial behavior. Even when entered last, following antisocial behavior, verbal ability, and family adversity, preschool ADD-H explained a full third of the models' explained variance in age 11 outcome (10%). In addition, examination of the partial correlations for the model's components revealed that age 5 ADD-H uniquely accounted for 12% of variance in age 11 antisocial behavior ($p < .001$).

Two of the model's interaction terms were modestly significant. Both family adversity and verbal ability may modify the risk for behavioral outcome that is posed by early ADD-H.

Prediction of future delinquency.—The last research question was whether or not the ADD + delinquent boys would be more delinquent than their peers when followed up at age 15. Based on the findings of previous research on prediction of criminal offending (reviewed by Farrington, 1987; Loeber & Stouthamer-Loeber, 1987; and Robins, 1979), it was hypothesized that the ADD + delinquent group, whose antisocial behavior was found to be characterized by (1) early onset and (2) persistent continuity, would be the group most likely to continue to engage in illegal behavior in the future. Although adult follow-up data are not yet available for the Dunedin cohort, self-reports of delinquency taken at age 15 could be examined (see Table 3). One-way analysis of group variance in age

15 SRED scores (which represented the variety of 29 illegal acts that each boy reported) yielded $F(3,418) = 13.28, p < .001$. Scheffé's comparisons between groups indicated that both delinquent groups reported significantly more different illegal behaviors at age 15 than nondisordered and pure ADD groups ($p < .05$). Of particular interest was the comparison between the ADD and non-ADD delinquents. These two groups did not differ significantly from each other ($p > .05$), although the ADD + delinquents reported marginally (.15 SD) more delinquent acts than the non-ADD delinquents.

Despite the fact that ADD and non-ADD delinquents reported that they had engaged in a similar number of different offenses at age 15, it was possible that the two groups' delinquent behaviors differed qualitatively. Therefore, the two delinquent groups were compared on each of the four categorical subscales of the SRED: aggression, theft, vandalism, and drug/alcohol-related offenses. The subscales are comprised of behaviors that are illegal, and the items were weighted for relative seriousness before summing (Moffitt & Silva, 1988b). The results are shown in Table 3. The ADD + delinquents' scores were highest on three of the four subscales, but only the aggressive subscale attained significance at conventional levels. Within-group variances were large. The ADD + delin-

quent boys reported a mean 1.5 higher SD aggression score than did the non-ADD delinquents. Age 15 is still within the middle adolescent period when delinquent behaviors are common for boys; the prediction that ADD + delinquents will continue to offend after temporary delinquents have desisted will be retested in later adolescence.

Discussion

Previous research with this and other samples has identified a subgroup of delinquents with a co-diagnosis of Attention Deficit Disorder. This group has characteristics that warrant closer investigation by developmental psychopathologists who are interested in the origins of antisocial behavior. In previous studies of this cohort, the ADD + delinquent boys were indistinguishable from the other ADD boys in hyperactivity and inattention, and they were indistinguishable from the other delinquent boys in amount of self-reported delinquency. As such, a study of either single behavior problem would probably not have uncovered their unique characteristics. The ADD + delinquent boys, when compared cross-sectionally to all other groups at age 13, suffered the greatest family adversity, had the worst neuropsychological deficits, and were rated as the most aggressive by their parents and teachers (Moffitt & Henry, 1989; Moffitt & Silva, 1988a), suggesting some discriminant validity for subgrouping by co-morbidity. The present study examined the developmental history of ADD + delinquency.

The results will be summarized by describing the findings for each study group in turn. The majority of the cohort, 80%, were nondisordered boys, whose scores on all measures provided a standard of comparison at the cohort mean at every age.

ADD nondelinquents.—The boys with ADD, but who were not delinquent at age 13, constituted about 4% of the cohort, and 46% of ADD cases. As a group, these boys evidenced no deficits in verbal IQ or motor abilities, nor any significant perinatal problems. They showed mild reading problems, but the difference was not statistically significant. Interestingly, their family adversity scores indicated that their home and family conditions were slightly more favorable than those of the cohort average. Their antisocial behavior tended to be mild and transient; they were significantly antisocial only in middle childhood (ages 9 and 11), and even then, the level of their antisocial behavior never approached that of their ADD + delinquent

peers. They had recovered from antisocial behavior by age 13, and by age 15 they reported slightly *less* involvement in vandalism and aggression than the cohort average.

The fluctuations seen in these boys' antisocial behavior, and their prosocial adolescent outcome, are encouraging because they indicate that much of their antisocial behavior is malleable, and therefore is probably susceptible to intervention. Their better-than-average family backgrounds and their good verbal intelligence may have contributed resilience to the risk posed by their Attention Deficit Disorder; even though they experienced some difficulty with reading, these boys did not become delinquent.

Non-ADD delinquents.—The boys who were designated as delinquents at age 13, but who had no prior history of ADD, made up 12% of the cohort, and 73% of all delinquents. The most interesting question about these boys is why they suddenly began to engage in delinquent behavior at the age of 13, after a childhood that was relatively free of antisocial behavior. Their antisocial behavior emerged at an accelerated rate after age 11, and by age 13 they had attained the same level of antisocial behavior shown by the ADD + delinquents. Non-ADD delinquents, as a group, had no significant deficit in verbal IQ, no difficulty with reading, no notable family adversity before age 11, and no perinatal or motor skills problems.

These boys developed delinquency in the absence of reading difficulty or low verbal IQ, indicating that theoretical explanations other than school failure are needed to account for their delinquency. Rutter (1979) has also reported that reading problems were not associated with conduct problems beginning in adolescence, as opposed to in early childhood. The delinquent behavior of these boys was less aggressive than that of their delinquent peers with ADD, and it may represent delinquency that is considered to be a "normative" and transient phenomenon of adolescence (Rutter & Garmezzy, 1983).

ADD + delinquents.—The ADD + delinquent group was small ($n = 19$); they represented about half of all ADD subjects, about 4% of the cohort, and one-quarter of the delinquents. Nevertheless, this group showed several notable developmental features. As a group, they began life with significant motor skills deficits and the greatest levels of family adversity. By the age of 5, they showed IQ deficits, which remained stable throughout development. These cognitive problems por-

tended difficulty in meeting the demands of school, and, indeed, these boys experienced reading failure soon after school entry and fell further behind their peers in reading as they approached high school. In the context of much fluctuation in antisocial behavior for the cohort as a whole, their antisocial behavior remained remarkably persistent. Their behavior deteriorated significantly over the years, a point which argues compellingly for very early intervention with ADD + aggressive boys.

The most striking increase in the antisocial behavior of ADD + delinquent boys occurred between the ages of 5 and 7, when they attained a mean antisocial rating that was not reached by the other delinquent boys until 6 years later. School entry and reading failure coincided temporally with this exacerbation of antisocial behavior. These data suggest that the problem behavior of this group, despite being generally persistent, *is* responsive to experience. The data also reveal a key point of vulnerability that could be a target for preventive intervention: reading readiness. Of course reading is not the only aspect of life that begins between ages 5 and 7. For example, sustained daily interaction with large groups of peers is also occasioned by school entry, and other researchers have noted that ADD + aggressive children experience special difficulty with processing social information in interpersonal interactions (Milich & Dodge, 1984).

The findings of this research suggest that the specific co-morbidity of ADD-H and antisocial behavior may offer discriminant validity in relation to some correlates of delinquency. In this sample, cases of co-morbidity between delinquency and other childhood disorders (anxiety and depression) were too few to perform analyses comparing various combinations of co-morbidity. When those few cases were examined, it appeared as if they scored differently from the delinquents with ADD co-morbidity and more similarly to controls on most of the study's measures. In addition, the relative rarity of overlap between "internalizing disorders" and delinquent behavior in this unselected representative sample suggests that disorders other than ADD may not be as important as ADD in the development of antisocial behavior (6% of the sample and 8% of delinquents had internalizing disorders, but 8% of the sample and 27% of delinquents had ADD). Nonetheless, further research of the specificity of ADD as a predictor of delinquent outcome is needed with larger samples of varied co-morbidities.

The data reported here might be taken as evidence that, given a sample of boys with ADD, differences in family environment make the difference in prognosis. Indeed, when a model of risk and vulnerability including age 5 ADD and family adversity was tested, the interaction of the two variables predicted a significant amount of the variance in antisocial behavior 6 years later. This interpretation would suggest that the differential prognosis for delinquent outcome may be entirely dependent upon environmental influences. However, two considerations advise that there is more to the prognosis of ADD boys than their family environments. These two considerations suggest an alternative, or complementary, biobehavioral hypothesis for the differential outcomes of the two ADD groups.

First, family adversity was not the only characteristic on which the two ADD groups differed. The group with the worst prognosis also had the poorest preschool motor performance and intelligence test performance. Intelligence test performance can be affected by social factors, of course, but it is unlikely that motor development measures reflect solely social factors. Some common central nervous system pathology may underlie their compounded problems of hyperactivity, inattention, aggression, poor motor skills, poor verbal abilities, and low reading achievement. The absence of that physiological pathology (together with family environment advantages) could be hypothesized to explain the relatively better prognosis of other ADD boys. In one study, Satterfield and Schell (1984) reported prospective group differences in amplitudes of the N2 component of the auditory event-related potential, suggesting that sub-optimal cortical arousal may have characterized ADD boys who became delinquent relative to those who did not. Hinshaw (1987) noted a paucity of research into the physiological, biochemical, and treatment-response characteristics of ADD subjects with and without antisocial behavior; more of this sort of work is needed.

Second, it is also the case that heritable traits shared by parents and their sons (such as poor verbal intelligence, physiological vulnerability to substance dependence, or a tendency toward behavioral dyscontrol) may contribute to the coexistence of behavior disorder in a child and adverse conditions in his family. The omnibus measure of family adversity used here was not a pure measure of *social* family environment. It contained elements, such as maternal IQ and maternal mental

health, that have documented heritable components (for reviews, see Bouchard & McGue, 1981; Cattell, 1982; Mednick, Moffitt, Pollock, Talovic, & Gabrielli, 1983), and that have been shown to relate to antisocial behavior (Moffitt et al., 1981; Moffitt, 1987). Thus, some heritable third variable may explain part of the correlation between family adversity and delinquent outcome for boys with ADD (Cantwell, 1972, 1975; Morrison & Stewart, 1971, 1973). Snyder and Patterson (1987) have called for research designed to disentangle the biological and social facets of family adversity, and the present results underscore this need.

It might also be hypothesized that boys from disadvantaged families were less likely to obtain treatment for their ADD, and that the differences in symptom picture and prognosis might therefore be ascribed to group differences in treatment histories. However, parents of ADD + delinquents were nearly four times more likely than parents of ADD-only boys to have sought help for their son from a psychologist, psychiatrist, family physician, teacher, or social worker by the age of 13. In this New Zealand sample, then, treatment seeking was more closely associated with the multiplicity of behavior problems than with advantaged family status.

In this study, family adversity and verbal intelligence were conceptualized as factors that afforded vulnerability to, or protection from, delinquency. Their interactions with ADD were found to relate significantly to antisocial behavior at age 11. One interpretation of that interaction is that strong verbal skills and/or good family circumstances can help to overcome risk for delinquency associated with ADD. Conversely, children who struggle under the combined burden of poor verbal intelligence and family adversity may be at marked risk for delinquent involvement as teens. The biological and environmental foundations of family adversity and verbal intelligence remained confounded in this design, but, regardless of origin, these two factors are clearly important targets for intervention research.

This research revealed that boys with the co-morbidity of ADD and delinquency showed antisocial behavior that began early in childhood and persisted to at least age 15, but the long-term outcome for these boys remains to be seen. In at least three other longitudinal studies, subjects with ADD and aggression who were followed up had continued their illegal behavior into young adult-

hood (Farrington et al., in press; Loney et al., 1983; Magnusson, 1988). Thus it appears that this subtype of childhood behavior may be especially likely to develop into adult crime. Early onset and persistence of antisocial behavior, aggressiveness, and engaging in a wide variety of illegal behaviors are recognized predictors of long-term criminal offending. Because the ADD + delinquent boys studied here appear to match those descriptors, it is anticipated that they will persist in delinquency beyond the time when the non-ADD delinquent group's "normative" delinquency desists. In this study, prediction from ADD status to delinquency at age 15 was obscured by the "noise" of normative delinquency. It is hypothesized that prediction from childhood ADD + antisocial behavior to later illegal behavior should improve as age increases (in violation of the "longitudinal law" that predictive accuracy tends to decrease as the time interval between observations increases). That hypothesis will be tested in future follow-ups of the sample.

This research has implications that bridge the realms of research methodology, theory, and intervention. The demonstration of distinct pathways to common presentations of delinquency suggests that research designs that fail to distinguish delinquent subgroups on the basis of co-morbidity may obscure important findings about the etiologies of delinquent behavior, and may be unable to predict adult outcomes accurately. In addition, because this study used biannual data collection waves, it was limited to a relatively telescopic look at the subjects' development; designs with more frequent assessments are desirable for fine-grained analysis of developmental trajectories. With respect to theory, the differing pathways found here hint that unitary theories of delinquency may not be sufficient to explain the possibly differing etiologies of "life-course persistent" and "adolescence-limited" antisocial behavior. Regarding intervention, this study illustrated that much childhood antisocial behavior does appear to be responsive to environmental influences, and it pointed to some factors that may be likely targets for preventive intervention. Finally, the findings point to the desirability of intervention that is truly early.

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